

Research Article

Highly active and controllable MOF-derived carbon nanosheets supported iron catalysts for Fischer-Tropsch synthesis



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ABSTRACT

N-doped carbon materials as catalyst supports have shown great superiority because of rich porosity, tunable metal-support interaction and electronic promotion effect. Controllable fabrication of carbon materials with specific morphology and different configurations provides wider possibilities for applications in catalysis. Herein, N-doped carbon nanosheets (CNS) were successfully synthesized by using monoclinic ZIF-8 nanosheets as templates. Compared with commercial active carbon and cubic ZIF-8 particles-derived carbon particles, the derived CNS exhibit superior graphitization degree and electronic effect. Consequently, the promoted Fe carburization improved the intrinsic activity for Fe/CNS catalysts in FTS. Further investigation on pyrolysis temperature achieved tunable configuration and content of N dopant. XPS, Raman, Mössbauer spectroscopy, TEM, H₂-TPR etc. were combined to elucidate the promotion effect of CNS on dispersion, electronic properties and formation of Fe carbides. By correlating the composition and structure to catalytic performance in FTS, we illustrated the significance of carbon materials in Fe supported catalysts.

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1. Introduction

Nanoporous carbon-based materials have been widely used in catalysis [1–3], adsorption [4,5], electrolytic capacitors [6] and fuel cells [7] due to their low cost, high surface area, rich porosity structure and good thermal stability [8,9]. Particularly, as catalytic supports, carbon materials usually exhibit weak metal-support interaction which favors metal reduction and enhances catalytic activity. However, metal species tend to migrate and aggregate during reaction, leading to a reduction in catalytic performance. Functionalizing carbon supports with heteroatom doping is known as one type of chemical modification strategy [10]. Recently, nitrogen-incorporated carbon materials have been widely studied in metal supported catalysts. The framework N atoms, as electronic donors, can enhance the electron density of adjacent metal species [11] and can help to anchor metal species by strengthening metal-support interactions, improving metal dispersion and sintering-

resistance [12,13]. Typically, different coordination environments and amounts of N dopants exhibit various and even surprising impacts in catalysis. Coville et al. [12] reported that pyridinic and pyrrolic N resulted in higher dispersion of Fe₃O₄ and consequently better activity, while graphitic N has little effect on activity. Li et al. [13] proposed that pyrrolic N generated more electron-enriched cobalt sites. By promoting CO adsorption and dissociation, an enhanced catalytic activity and a higher C₅₊ selectivity was achieved. In contrast, Duan and Zhou's group [14] reported that pyridinic and pyrrolic N decreased the electron density of Pd atoms, while the presence of graphitic N caused more electron-enriched Pd sites and accelerated hydrogenation of acetylene.

Fischer-Tropsch synthesis (FTS) is a sustainable process to produce various valuable chemicals derived from non-petroleum sources (i.e. syngas) [15,16]. Carbon materials are considered good support candidates for FTS due to their high surface areas and weak metal-support interaction, which favors dispersion, reduction and carburization of Fe species [17]. Textural properties of carbon materials, such as morphology, porosity, and regularity etc., usually play important roles in performance. Although N doped carbon materials as supports have been reported to improve FTS

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performance [18,19], identifying the role of N species is still of great significance for further catalyst design and modification.

Metal organic frameworks (MOFs) have been considered to be an ideal sacrificial precursor for fabricating heteroatom-doped porous carbon materials [20,21], due to their abundant porosity and tunable functional organic ligands. As a typical MOF, zeolitic imidazolate framework (ZIF-8) has gained much attention because complete evaporation of Zn could be achieved when the pyrolysis temperature is above 900 °C [22]. The content and configuration of N can also be controlled by adjusting the pyrolysis temperature [23], based on thermal stability difference of different N species. Furthermore, ZIF-8 has changeable crystal morphologies (e.g. cube and nanosheet) [24], which provides an opportunity to investigate the morphology effect on carburization and catalytic performance.

In this study, we employed monoclinic ZIF-8 derived carbon nanosheets as supports for preparing Fe catalysts (Fig. 1). Compared with commercial active carbon and cubic ZIF-8 derived carbon particles, we aimed to investigate the effect of morphology and N dopant on FTS performance. The unique two-dimensional structure together with N-dopant resulted in iron catalysts with excellent activity and stability in FTS at high temperature. By further controlling the carbonization temperature of MOF precursors, we successfully regulated the nature and content of N species, which were proved to have considerable impacts on FTS performance.

2. Experimental section

2.1. Catalysts preparation

Synthesis of the carbon nanosheets supports (CNS) and Fe/CNS.

The N-doped carbon materials as supports were synthesized by pyrolysis of N-containing MOF (i.e. ZIF-8). Firstly, 0.33 g $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (CAS No: 10196-18-6, purity: 98%, Shanghai Aladdin Bio-Chem Technology Co., Ltd) and 0.99 g 2-methylimidazole (CAS No: 693-98-1, purity: 98%, Shanghai Aladdin Bio-Chem Technology Co., Ltd) were dissolved into 90 mL deionized water separately, then mixed together and magnetically stirred for 24 h at room temperature. The white precipitate was separated by centrifugation, followed by washing and drying in air at 50 °C for 12 h and was denoted as ZIF-8-nanosheets (ZIF-8-NS). Then the ZIF-8-NS was pyrolyzed at 900, 950, 1000 and 1100 °C for 2 h with a rate of 2 °C/min under argon flow (120 mL/min) in a tube furnace. The carbonization products were marked as CNS(T), where 'T' stands for the calcination temperature.

The catalysts with nominal 6.5 wt% iron loading were prepared using incipient wetness impregnation. 0.214 g of ammonium iron citrate (CAS No: 1185-57-5, purity: AR, Tianjin Guangfu Fine Chemical Research Institute) was firstly dissolved in 1 mL deionized

water, and impregnated inside the pores of the obtained CNS under stirring, followed by ultrasonic treatment. The resultant materials were dried overnight at 50 °C under vacuum and calcined under argon flow (120 mL/min) at 500 °C for 2 h with a heating rate of 5 °C/min in a tube furnace. Subsequently, the samples were passivated by a gaseous mixture of 1 vol% O_2 in Ar for 1 h. The final products were named as Fe/CNS(T).

Synthesis of Fe/CP and Fe/AC. For comparison, cubic ZIF-8 particles (ZIF-8-P), mediated carbon particles (CP) and commercial active carbon (AC) were also used as supports. The CP was obtained by calcining the commercial ZIF-8-P (Jiangsu XFANANO Materials Tech. Co., Ltd) at 1000 °C for 2 h with a rate of 2 °C/min under argon flow (120 mL/min). The AC (Fujian XINSEN Carbon Co., Ltd) was pre-treated by immersing into an HF solution to remove impurities, and then drying and calcining under same conditions. The Fe/CP and Fe/AC catalysts were also prepared by incipient wetness impregnation.

2.2. Catalyst characterizations

X-ray powder diffraction (XRD) patterns were measured on a Rigaku D/max-2500 X-ray diffractometer at 40 kV and 200 mA with the Cu K_α radiation. The spectra were scanned with a speed at 5°/min in the 2θ range of 10°–90°.

Nitrogen adsorption isotherms were recorded at –196 °C on Micromeritics ASAP 2460 instrument. Prior to testing, the samples were evacuated at 300 °C for 4 h to remove the water and impurities. The surface areas of samples were obtained from the Brunauer-Emmett-Teller (BET) method. Their total pore volumes were calculated based on the adsorption amount of nitrogen at $P/P_0 = 0.95$ and their corresponding micropore volumes were determined by the t-plot method. The pore size distribution was obtained by using the NLDFT method.

Scanning electron microscope (SEM) was performed with an S-4800 field-emission microscope (Hitachi, Japan) at an accelerating voltage of 5 kV.

Transmission electron microscope (TEM) was carried out to observe the morphology and particle dispersion of samples on a Philips TESNAI G2 F20 system microscope with the electron accelerating voltage of 100 kV. The particle size distribution was calculated by more than 300 nanoparticles at different regions. The EDX-mapping images were acquired in the dark field to analyze the element dispersion.

The Raman spectra were recorded on a Raman spectrometer system (LabRAM HR Evolution, HORIBA Scientific) equipped with a laser excitation of 532 nm. The resolution is 1.838 cm^{-1} .

X-ray photoelectron spectroscopy (XPS) measurements were performed with a ThermoFischer Scientific spectrometer with an Al K_α excitation source. The binding energies were calibrated by

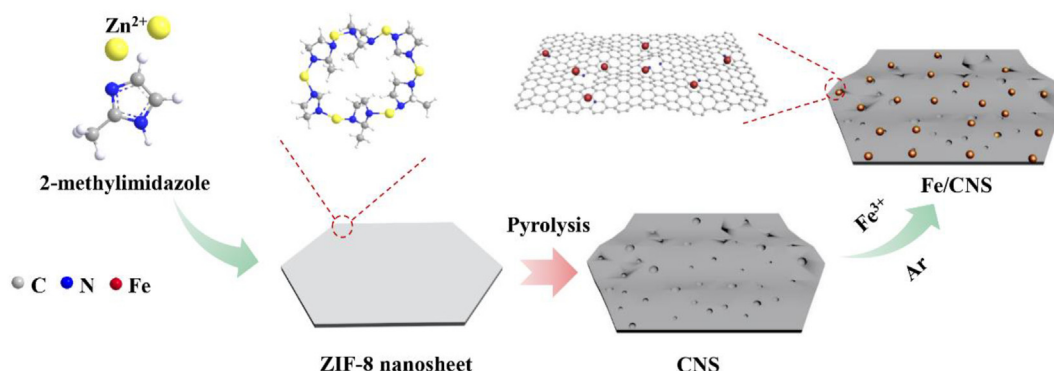


Fig. 1. Schematic illustration of the synthesis process for preparing Fe/CNS catalyst. (A colour version of this figure can be viewed online.)

choosing C 1s peak (284.8 eV) as a reference. All samples were tested in vacuum with 5×10^{-8} Pa.

H₂ temperature programmed reduction (H₂-TPR) were performed using a Micromeritics AutoChem 2910 equipped with a thermal conductivity detector (TCD). 100 mg of samples were pretreated in an Ar atmosphere at 200 °C for 1 h, followed by cooling to room temperature. Then, the gas mixture of 10% H₂ and 90% Ar was injected into the system and the H₂ consumption was continuously detected from 50 to 600 °C.

Temperature-programmed desorption of CO (CO-TPD) was carried out on a Micromeritics Autochem II 2920 chemisorption apparatus. The catalysts were reduced at 350 °C in an H₂ flow for 2 h and exposed to 10% CO in Ar at room temperature for 1 h. After He purging for 1 h, the desorbed CO was monitored by a mass spectrometer (m/z of 28 for CO) from 25 to 900 °C.

The room temperature ⁵⁷Fe Mössbauer spectra were recorded using a proportional counter and a Topologic 500A spectrometer. A ⁵⁷Co(Rh), moving with a constant acceleration mode, was used as the γ -ray radioactive source. The velocity was calibrated by a standard α -Fe foil. The spectra were fitted on the basis of Lorentzian adsorption curves using MossWinn 3.0i computer program. The derived hyperfine parameters, such as isomer shift (IS), quadruple splitting (QS), and magnetic hyperfine field (H), were employed for component identification. The phase content was obtained based on the areas of the adsorption peaks, assuming the iron nuclei for all catalysts has the same probability of adsorption of γ photons. For the measurement of all spent catalysts, we used the samples after 30 h reaction.

2.3. Activity tests

The FTS testing was conducted in a tubular fixed-bed reactor with an inner diameter of 8.0 mm. Typically, 100 mg of catalyst was evaluated at 340 °C and 1 MPa. The catalysts were activated in situ with a flow of H₂ (15 mL/min) at 350 °C and ambient pressure for 4 h. Then the temperature was decreased to the reaction temperature, and feed gas was controlled to flow into the reactor with a certain flow rate. The feed gas contained 45% CO, 45% H₂ and 10% Ar in volume ratio and Ar acted as an internal standard gas. The high carbon hydrocarbon products were collected in a hot trap at 135 °C. The residual feed gas and light products (oxygenates, C₁–C₆ hydrocarbons, CO₂) were analyzed online with an Agilent 7890B gas chromatograph equipped with a flame ionization detector (FID) and thermal conductivity detector (TCD). Separation was achieved using two columns: an UltiMetal column for permanent gases, CH₄ and CO₂, and an HP-AL/S column for oxygenates and hydrocarbon gases, which were separately detected with the TCD and FID detector. The liquid products in the hot trap were collected once per 20 h. The carbon balance on Fe/AC, Fe/CP and Fe/CNS(1000) from 10 to 30 h changes from 95.0 to 95.9%, because of gradual carburization of Fe species. For Fe/CNS(T) series, the carbon balance is higher than 96.5% after 30 h.

The Fe time yield (FTY) was determined as following:

$$FTY = \frac{\text{Moles of CO converted to hydrocarbons}}{\text{Mass of Fe atoms} \times \text{Time}}$$

3. Results and discussion

3.1. Physicochemical properties and performance of Fe/CNS, Fe/CP and Fe/AC

3.1.1. Morphology and structure of catalysts

With the aim to investigate the effect of morphology on FTS, we

compared ZIF-8-nanosheets (ZIF-8-NS) and commercial ZIF-8-particles (ZIF-8-P) as support precursors. The SEM images (Fig. 2a and b) show that the ZIF-8-NS exhibits a typical sheet-like shape with a thickness of 30–50 nm, while ZIF-8-P was found to have a cubic morphology with diameters of ~200 nm. The crystal diffraction peaks of ZIF-8-NS and ZIF-8-P (Fig. 2c) are consistent with those reported characteristics for monoclinic and cubic ZIF-8 [24], confirming the successful preparation of the ZIF-8 with specific crystal phases. After carbonization and Fe loading, both nanosheet and cubic morphologies were almost preserved, as shown in Fig. 2d, g and e, h. However, the AC and Fe/AC possess an irregular and porous network structure, shown in Fig. 2f, i. All XRD patterns of the three supports have two broad peaks at 24–26° and 44° (Fig. 3a), which are assigned to (002) and (101) diffractions of carbon materials. The complete disappearance of ZIF-8 features and absence of zinc related peaks on CNS(1000) and CP suggest framework collapse, evaporation of zinc and the formation of carbon matrix at high temperature [24,25]. The diffraction peaks at around 25° of CNS(1000) and Fe/CNS(1000) are sharper and narrower, shifting towards to larger range of 2θ (Fig. 3a and b). These observations indicate that the crystal and morphological features of ZIF-8 affect the degree of graphitization of the obtained carbon materials. The ZIF-8-NS tends to form an ordered carbon structure [26].

Raman spectroscopy was further used to reveal the graphitization degree of the three carbon supports (Fig. 3c). All catalysts display two prominent peaks at approximately 1340 and 1580 cm⁻¹, corresponding to typical D-band and G-band of carbon materials [27–29]. The intensity ratio of D to G band (I_D/I_G) is usually used as a descriptor to reflect the defect and disordered level. The CNS(1000) shows a lower value of I_D/I_G than that of CP and AC, suggesting its higher graphitization degree, which is in agreement with the XRD results. After loading iron, the reflections related to iron phases in the XRD patterns are observed (Fig. 3b). The peaks at 30.1°, 35.4°, 43.0°, 57.0° and 62.5° are attributed to (220), (311), (400), (511) and (440) planes of Fe₃O₄ (JCPDS 19–0629) [30], whereas the peaks at 44.7° and 82.3° correspond to the (110) and (211) facets of Fe (JCPDS 65–4899) [31]. The broad weak swell at 41–43° suggest the possible presence of iron carbides (FeC_x). Due to the overlap of characteristics of different FeC_x species, it is hard to identify the precise crystal structure of Fe carbides in these catalysts [32]. The surface carbon and residual organic components of supports may function as reduction and carburization agents under an Ar atmosphere, yielding a small amount of Fe and FeC_x, which is consistent with our group's findings [33,34]. Fig. S1 shows that the Mössbauer spectra of fresh Fe/AC, Fe/CP and Fe/CNS(1000) catalysts all consist of one doublet and three sextets. According to the results summarized in Table S1, Fe species in all fresh catalysts mainly exist as Fe(II)/Fe(III) together with small proportion of χ -Fe₅C₂ [35,36]. Besides, comparing to AC, ZIF-8 derived carbon supports facilitate formation of χ -Fe₅C₂ even before exposure to syngas.

N₂ physical adsorption (Fig. 4, Table 1) indicates that ZIF-8 as a precursor provides large surface areas and abundant micro- and meso-porosity. The porous structure of the carbon materials results in better dispersion of Fe species. A reduction in surface areas and pore volumes after Fe loading indicates Fe species located in the channels. CNS(1000) exhibits significant reduction of pore volume and surface area after Fe loading, in comparison with CP and AC. The phenomenon is more visible in pore distribution curves (Fig. 4b, d). This infers that Fe precursors are prone to enter and remain in the pores of CNS(1000), probably benefiting from its 2D morphology feature as well as interaction of Fe and N atoms.

TEM images of fresh catalysts (Fig. 3d–f) reveal that the average particle size of Fe/CNS(1000) and Fe/CP are almost the same

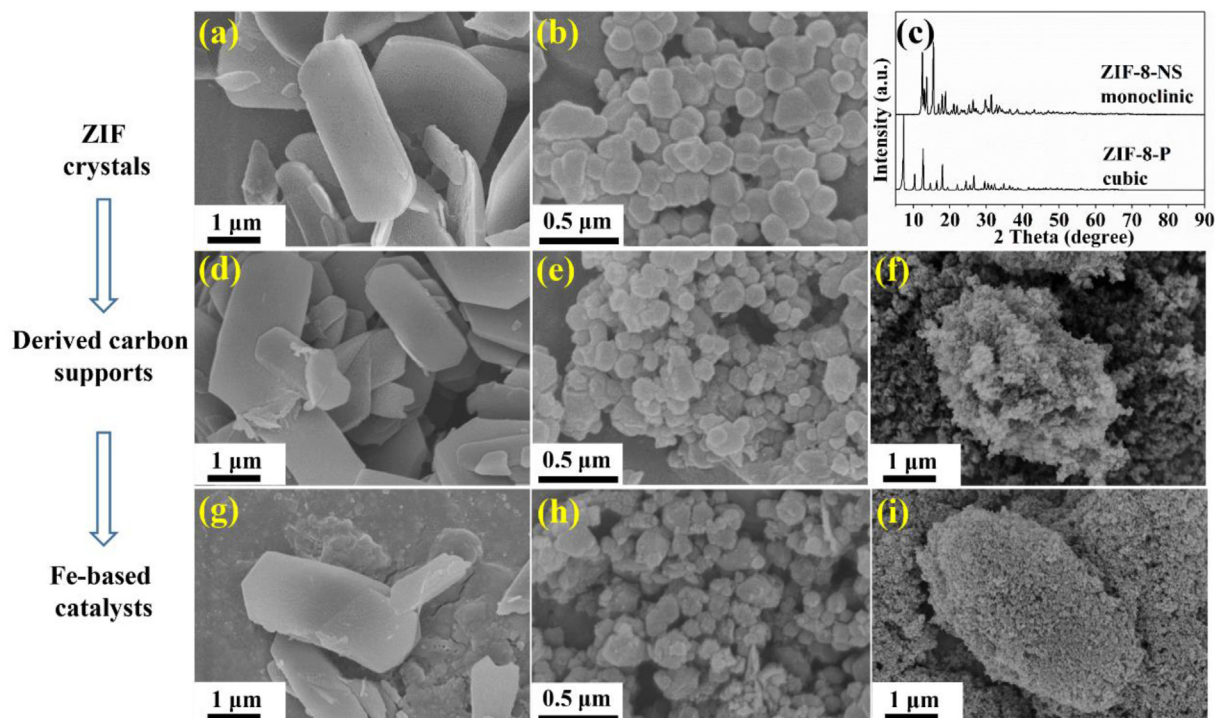


Fig. 2. SEM images (a–b) and XRD patterns (c) of ZIF crystals: ZIF-8-NS and ZIF-8-P; SEM images of derived carbon supports (d–f) and corresponding Fe-based catalysts (g–i): CNS(1000), CP, AC and Fe/CNS(1000), Fe/CP, Fe/AC. (A colour version of this figure can be viewed online.)

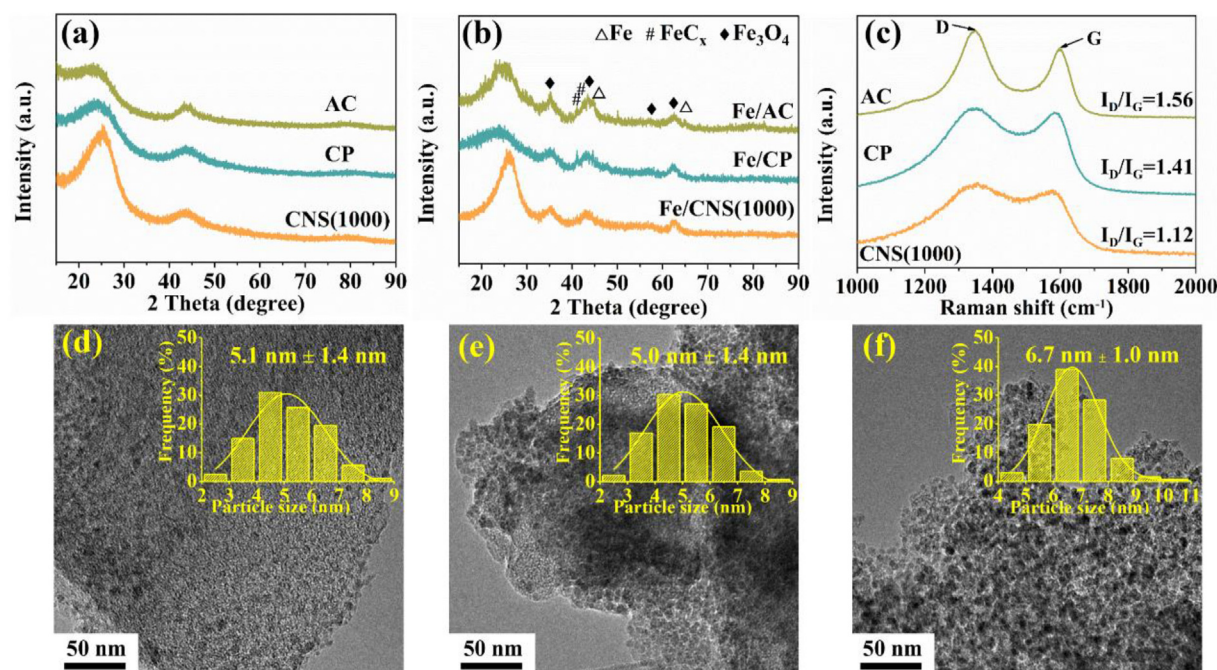


Fig. 3. XRD diffractions of different carbon supports (a) and corresponding Fe-based catalysts (b); Raman spectra of the supports (c); TEM images of Fe/CNS(1000) (d), Fe/CP (e) and Fe/AC (f) (inserted: size distribution of Fe nanoparticles). (A colour version of this figure can be viewed online.)

(~5.0 nm) but smaller than that of Fe/AC (~6.7 nm). For nitrogen-containing MOF mediated carbons, the N atoms are uniformly dispersed after pyrolysis, which is evidenced by EDX-mapping in Fig. S2. The discrepancy in Fe dispersion for different catalysts is probably due to the existence of N atoms in the carbon matrix, which were reported as anchors to trap and stabilize metal species [13].

XPS was further applied to get an insight into the N configuration for the carbon-supported catalysts. In Fig. 5a, the overall XPS spectra demonstrate the presence of N element in Fe/CNS(1000) and Fe/CP, whereas Fe/AC is a N-free sample. This confirms the formation of N-doped carbon structures by pyrolyzing ZIF-8 with different morphologies [37–39]. The N 1s spectra of Fe/CNS(1000)

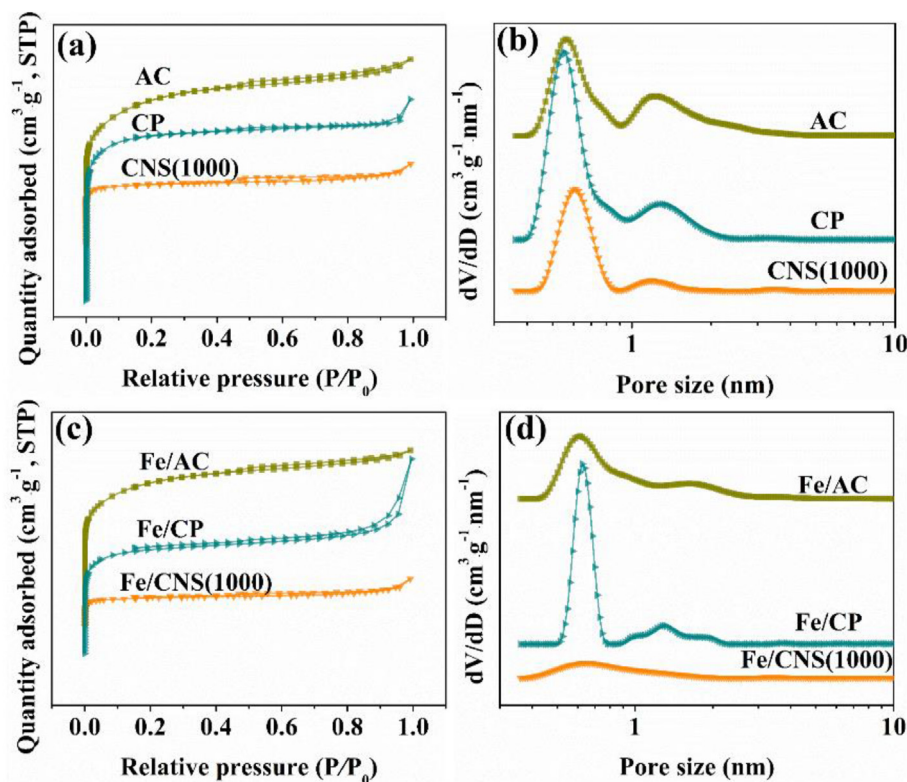


Fig. 4. N_2 adsorption-desorption isotherms (a, c) and pore size distribution (b, d) of the CNS(1000), CP and AC supports and corresponding iron-based catalysts. (A colour version of this figure can be viewed online.)

Table 1

Surface area and pore structure parameters of AC, CP, CNS(1000) and corresponding catalysts.

samples	S_{BET} ($m^2 g^{-1}$)	V_{total} ($cm^3 g^{-1}$)	D_a (nm)
AC	951	0.49	2.1
CP	1131	0.49	1.7
CNS(1000)	438	0.20	1.8
Fe/AC	769	0.40	2.1
Fe/CP	918	0.41	2.1
Fe/CNS(1000)	210	0.14	2.1

and Fe/CP could be deconvoluted into four peaks, as shown in Fig. 5b, representing the bond states of pyridinic N (N1, ~398.6 eV),

pyrrolic N (N2, ~400.1 eV), graphitic N (N3, ~401.1 eV) and oxidized N (N4, ~403.1 eV), respectively [40–43]. The total surface N and various N containing species in Fe/CNS(1000) and Fe/CP are almost identical (Table 2), revealing that the morphology of ZIF-8 has negligible impact on composition of N containing species at the same carbonization temperature.

As a response to local electronic environments, electron density changes in outer core level (Fe 3p) are more sensitive in comparison to inner core level (Fe 2p) [44–46]. Thus, Fe 3p spectra were chosen to analyze the changes in electronic density of Fe species. As shown in Fig. 6a, the binding energies (BEs) of Fe 3p spectra for Fe/CNS(1000) and Fe/CP shift to lower values compared with that of Fe/AC. This indicates that N acts as an electron-donor [47],

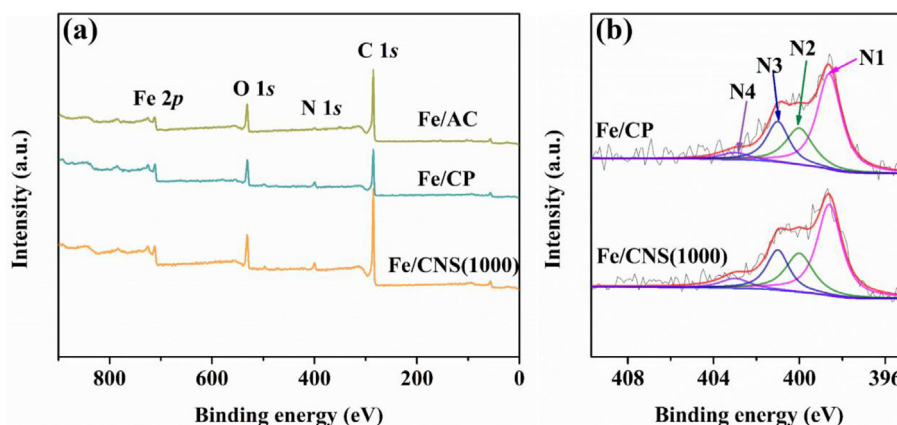


Fig. 5. Overall XPS spectra (a), and N 1s spectra (b) for Fe/AC, Fe/CP and Fe/CNS(1000) (N1, pyridinic N; N2, pyrrolic N; N3, graphitic N; N4, oxidized N). (A colour version of this figure can be viewed online.)

Table 2
Content of various N species and CO uptake amount for Fe/AC, Fe/CP and Fe/CNS(1000).

Catalysts	surface total N (atom %)	oxidized N (atom %)	graphitic N (atom %)	pyrrolic N (atom %)	pyridinic N (atom %)	CO uptake ($\mu\text{mol}_{\text{CO}} \text{g}_{\text{cat}}^{-1}$)
Fe/AC	—	—	—	—	—	208
Fe/CP	5.0	0.5	1.2	1.2	2.1	283
Fe/CNS(1000)	4.9	0.4	1.2	1.3	2.0	368

enhancing the negative charge density of Fe species. Even with similar configuration and content of N species to Fe/CP, the Fe/CNS(1000) exhibits a lower value of Fe 3p BE, demonstrating that a proper graphitization degree could contribute to the electron transfer and further affect electron density of the adjacent Fe atoms.

Considering CO adsorption and activation on active sites is a prerequisite in FTS, CO-TPD was conducted to analyze the CO adsorption behavior over the three samples. To exclude the decomposition and release of residual organic components of MOF-derived carbon materials with increasing temperature as well as the interaction between CO and MOF-mediated supports, CNS(1000) was performed first after CO pre-adsorption. In Fig. 6b, no CO signal was detected, demonstrating that CNS(1000) shows enhanced thermal stability even at 900 °C and that there is no active site that is capable of interacting with CO on derived carbon supports. For Fe/CNS(1000) and Fe/CP, two CO peaks at around 500–580 °C and 700–720 °C were observed, which are ascribed to desorption of moderately and strongly adsorbed CO, respectively. In contrast, there is only a single peak located at around 610 °C for Fe/AC. The amount of adsorbed CO increases in the order of Fe/CNS(1000) > Fe/CP > Fe/AC (Table 2). As evidenced by XPS, N doping and higher graphitization facilitate electron transfer and donation from support to Fe species, resulting in a decline in binding energies of Fe 3p. Therefore, the enhanced CO adsorption capacity can be attributed to the two factors, which is in line with Hu's work [28].

3.1.2. Catalytic performance of catalysts

To verify the support effect of different carbon materials on FTS, Fe/CNS(1000), Fe/CP and Fe/AC were evaluated. As listed in Table 3, Fe/AC displays the lowest CO conversion of 7.7%, while activities of MOF-derived N doping carbon supported Fe catalysts (i.e. Fe/CP and Fe/CNS(1000)) are dramatically improved to 22.1% and 45.9%. This demonstrates the crucial role of carbon supports in FTS performance. The surface area and pore volume of Fe/AC and Fe/CP are similar, but much higher than those of Fe/CNS(1000). This divergence indicates that the difference in activities are not caused by physical properties. At the similar graphitization degrees (i.e. Fe/CP and Fe/AC), the electron donation of N in the Fe/CP strengthens Fe–C bond and weakens C–O bond when CO is adsorbed, which is

due to the enhanced back-donation of Fe valence electrons into the π -orbital of C in CO [46]. Consequently, the CO conversion is improved. On the other hand, the FTY on Fe/CNS(1000) is nearly two times greater than that for Fe/CP, based on similar nitrogen content and nature. CO-TPD curves suggested that moderate strength and improved capacity of CO adsorption for Fe/CNS(1000) can benefit catalytic performance in FTS, consistent with Lok's findings [48]. The phase compositions of the used Fe/AC, Fe/CP and Fe/CNS(1000) catalysts were further determined by the Mössbauer spectra (Fig. 7 and Table 4). A large number of χ -Fe₅C₂ species are formed under the syngas atmosphere. It is noteworthy that the content of carbides in used catalysts follows the order: Fe/CNS(1000) > Fe/CP > Fe/AC. Correspondingly, the FTY values follow the same trend. By analyzing product selectivity, we found that N-doped catalysts (i.e. Fe/CP and Fe/CNS(1000)) show lower C₅₊ product selectivity and higher CH₄ selectivity compared with Fe/AC. The trend became more pronounced at similar CO conversions (see Table 3). Therefore, both morphology and N-doping affects CO conversion significantly, while N-doping affects products distribution.

The stabilities of the catalysts are shown in Fig. 8a. Fe/AC exhibits the shortest lifetime, which deactivates after 15 h on stream. Considering data reliability at extremely low CO conversion, we listed their performance data at TOS = 15 h for comparison (Table 3). Benefiting from the anchoring effect of N, Fe/CNS(1000) and Fe/CP show superior stability. Note that Fe/CNS(1000) shows an obvious activation stage, which was caused by gradual carburization with reaction time. Further comparison of TEM images (Fig. 8b–d) reveals that the used Fe/AC catalyst exhibits severe particle agglomeration. This is also supported by XRD patterns of spent catalysts (Fig. S7), in which the used Fe/AC(1000) shows sharp diffraction peaks of Fe₅C₂ crystal phase. This proves that the N dopant as the anchoring site is vital to improve catalyst stability.

To summarize, MOF-derived N-doped carbon with nanosheet morphology as a support shows great advantages in FTS, owing to its special morphology and the electronic effect of N species. In the next section, we further tuned the carbonization temperature to elucidate the contents and nature of N dopants effect on catalytic performance.

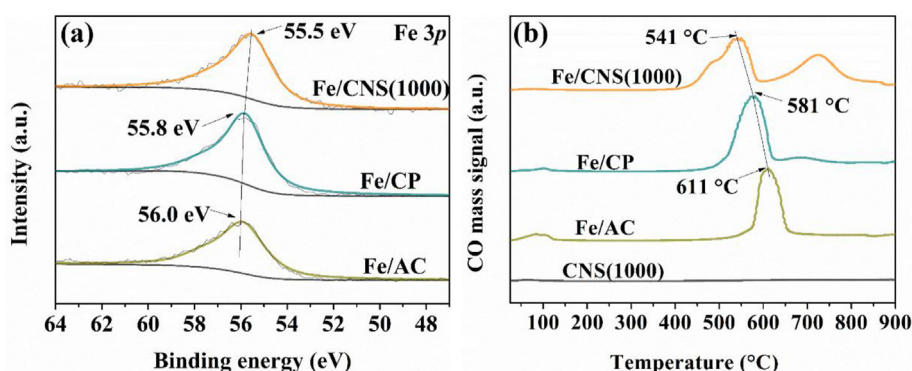


Fig. 6. Fe 3p spectra (a) and CO-TPD profiles (b) of Fe/AC, Fe/CP and Fe/CNS(1000). (A colour version of this figure can be viewed online.)

Table 3
FTS performance of Fe/AC, Fe/CP and Fe/CNS(1000) catalysts.

Catalysts	CO conv. (%)	FTY ($\mu\text{mol}_{\text{CO}} \text{g}_{\text{Fe}}^{-1} \text{s}^{-1}$)	CO ₂ selectivity (%)	Hydrocarbon selectivity (C-mol%)					O/P
				CH ₄	C ₂ –C ₄	C ₂ –C ₄	C ₅ +	Oxy.	
Fe/AC ^a	7.7	57	13.0	13.1	14.3	9.7	62.3	0.6	1.5
Fe/CP ^a	22.1	144	24.2	23.6	21.6	16.6	37.2	1.0	1.3
Fe/CNS(1000) ^a	45.9	278	29.4	22.4	20.5	16.9	39.2	1.0	1.2
Fe/CP ^b	10.1	131	21.7	27.2	31.2	12.5	28.5	0.6	2.5
Fe/CNS(1000) ^c	10.2	269	22.9	35.0	37.1	11.2	16.2	0.5	3.3

Reaction conditions: 0.1 g cat., 340 °C, 1 MPa, H₂/CO = 1, TOS = 15 h, ^a GHSV = 9000 mL g^{−1} h^{−1}, ^b GHSV = 18000 mL g^{−1} h^{−1}, ^c GHSV = 36000 mL g^{−1} h^{−1}.

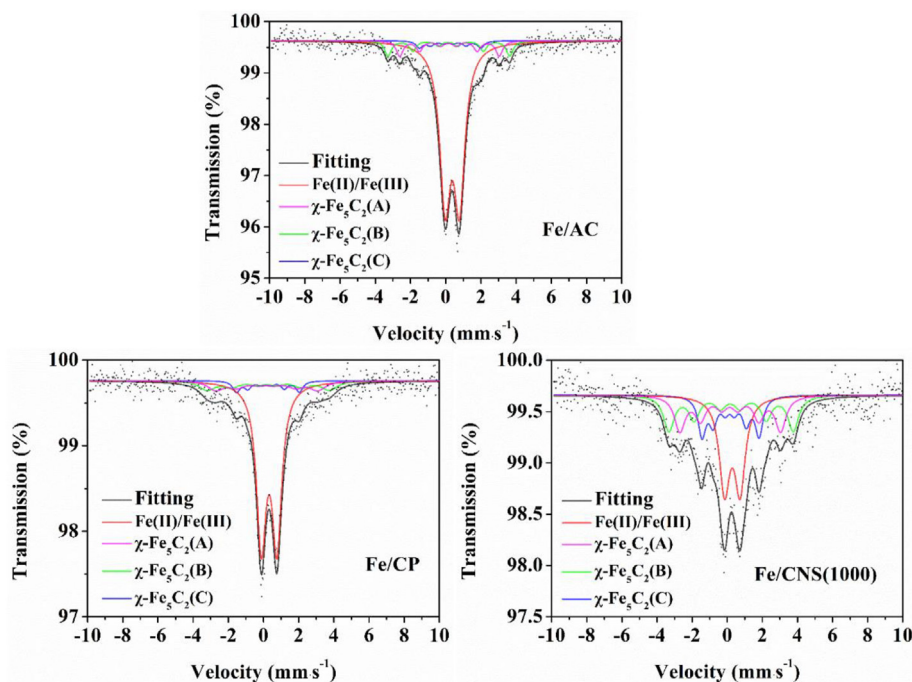


Fig. 7. ⁵⁷Fe Mössbauer spectra of the used Fe/AC, Fe/CP and Fe/CNS(1000) catalyst. (A colour version of this figure can be viewed online.)

Table 4
⁵⁷Fe Mössbauer parameters of the used Fe/AC, Fe/CP and Fe/CNS(1000) catalysts.

Used catalysts	IS (mm·s ^{−1})	QS (mm·s ^{−1})	H (T)	Phase ascription	A (%)
Fe/AC	0.38	0.81	—	Fe(II)/Fe(III)	76.5
	0.18	0.06	17.6	χ-Fe ₅ C ₂ (A)	9.5
	0.16	0.04	21.5	χ-Fe ₅ C ₂ (B)	9.6
	0.20	0.11	10.8	χ-Fe ₅ C ₂ (C)	4.4
Fe/CP	0.32	0.88	—	Fe(II)/Fe(III)	68.2
	0.17	0.03	18.0	χ-Fe ₅ C ₂ (A)	12.4
	0.22	0.09	21.6	χ-Fe ₅ C ₂ (B)	12.5
	0.20	0.11	11.1	χ-Fe ₅ C ₂ (C)	6.9
Fe/CNS(1000)	0.29	0.89	—	Fe(II)/Fe(III)	31.2
	0.15	0.03	17.8	χ-Fe ₅ C ₂ (A)	24.4
	0.20	0.05	22.0	χ-Fe ₅ C ₂ (B)	24.5
	0.16	0.05	10.1	χ-Fe ₅ C ₂ (C)	19.9

3.2. Tuning effect of N dopants on FTS performance

3.2.1. Structure of Fe/CNS(T) catalysts

Carbonization of ZIF-8 nanosheet at different temperatures (900–1100 °C) was conducted. XRD patterns of Fe/CNS(T) catalysts (Fig. 9a) evidence the co-existence of three iron phases; Fe, FeC_x and Fe₃O₄. The Fe/CNS(900–1000) samples exhibit relatively low peaks in the range of 41°–47°, while the diffraction peaks of FeC_x for Fe/CNS(1100) are more easily distinguished, which is indicative

of higher crystallinity. In addition, no zinc related species are observed, suggesting its complete sublimation at temperatures higher than 900 °C [49]. As the carbonization temperature increases, diffraction of (002) facet becomes narrower and more intense, indicating gradually improved graphitization degree, which is also verified by decreased values of I_D/I_G in the Raman spectra (Fig. 9b). However, the changes in graphitization level for Fe/CNS(T) are relatively moderate, in comparison with the samples derived from different precursors. This implies that precursor property is a crucial factor to influence the derived carbon materials.

As shown in Table 5 and Fig. 10a, XPS results suggest that the total amount of surface N atoms gradually decreases from 7.8% to 3.8% with increasing carbonization temperature. Severe decomposition of organic ligands occurs on CNS(T) at higher pyrolysis temperatures, accompanied by the removal of N elements [50] and enriched porosity (Fig. S3 and Table S2). The dramatic decrease of adsorption quantity at around 0.6 nm confirms occupation of Fe species in micropores. To some extent, the confinement effect of microporous structure inhibits particle aggregation and growth during reaction. Deconvolution of N 1s spectra provides quantitative information about N with different configurations (Fig. 10b). With elevating the carbonization temperature, the ratio of pyridinic N tends to decrease, while graphitic N increases simultaneously (Fig. 10c). The relative content of pyrrolic N firstly increases then

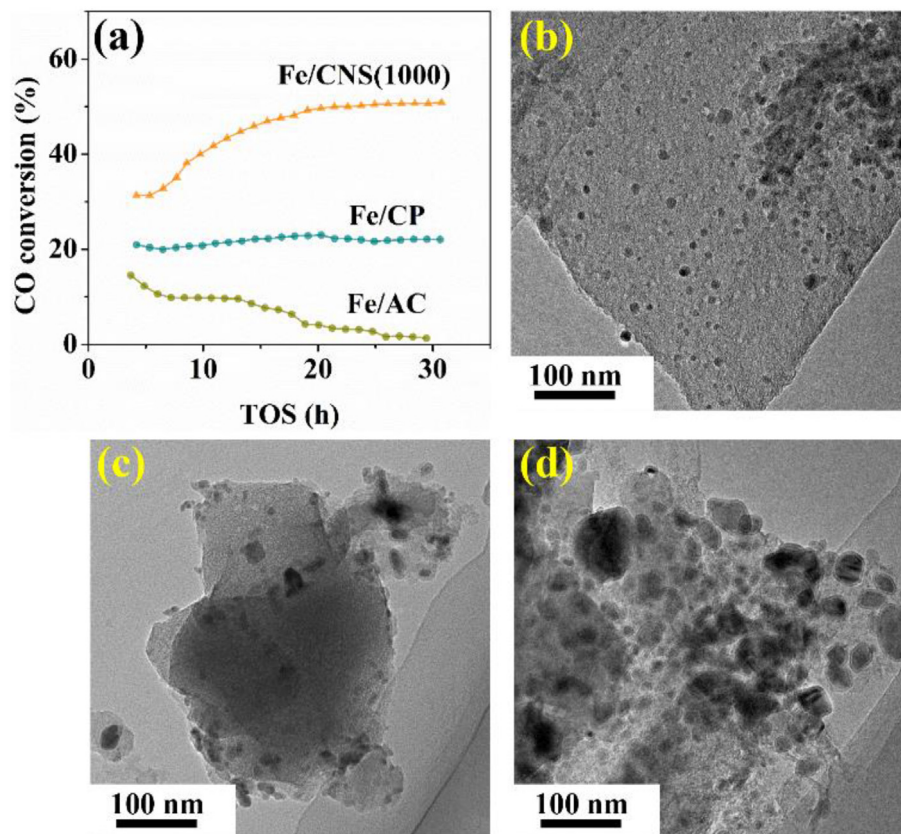


Fig. 8. CO conversions on TOS (a); and TEM images (b–d) of the used Fe/CNS(1000), Fe/CP and Fe/AC. (A colour version of this figure can be viewed online.)

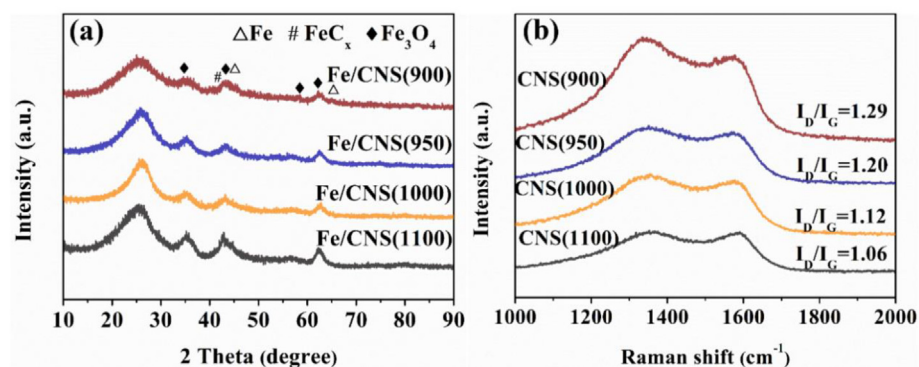


Fig. 9. XRD patterns (a) and Raman spectra (b) of Fe/CNS(T) catalysts. (A colour version of this figure can be viewed online.)

Table 5

Content of various N species for Fe/CNS(T) catalysts.

Catalysts	surface total N (atom %)	oxidized N (atom %)	graphitic N (atom %)	pyrrolic N (atom %)	pyridinic N (atom %)
Fe/CNS(900)	7.8	0.5	1.1	0.9	5.3
Fe/CNS(950)	5.9	0.5	1.1	1.1	3.2
Fe/CNS(1000)	4.9	0.4	1.2	1.3	2.0
Fe/CNS(1100)	3.8	0.4	1.2	0.8	1.4

starts to decline slightly after 1000 °C. Considering the changes in the total content of N with temperature, we also summarized the actual content of different N species for Fe/CNS(T) in Table 5.

To further study the interaction between Fe and N, H₂-TPR of Fe/CNS(T) was performed. Two peaks of H₂ consumption are

presented in Fig. 11, which can be assigned to the reduction of Fe₃O₄ → FeO [51] and FeO → Fe [52]. The two reduction peaks shift towards lower temperature with increasing pyrolysis temperature, indicating enhanced reducibility of Fe oxides. This indicates that the interaction between Fe and N was successfully tailored by

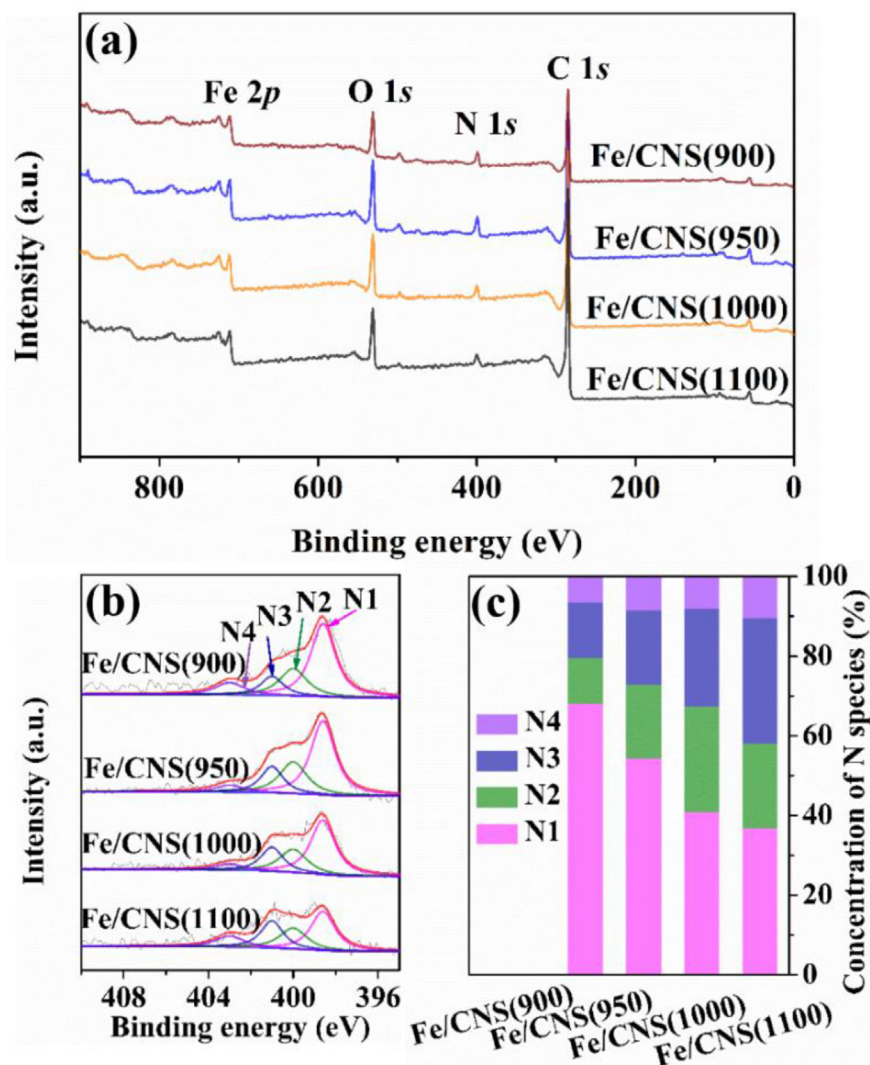


Fig. 10. Overall XPS spectra (a), N 1s spectra (b) and relative concentration of various N species (c) for Fe/CNS(T) (N1, pyridinic N; N2, pyrrolic N; N3, graphitic N; N4, oxidized N). (A colour version of this figure can be viewed online.)

changing the calcination temperature [53]. However, too weak interaction results in a limited anchoring effect for iron species, which is confirmed by TEM images of Fe/CNS(T) catalysts (Fig. 12). The average particle size of iron species on Fe/CNS(900–1000) almost retains the same size, with a size of ~5.0 nm. In contrast, the Fe/CNS(1100) possesses a larger particle size of ~8.9 nm, consistent with more intense diffractions in the XRD pattern (Fig. 9a). In short, an appropriate interaction between Fe and N is beneficial for both reduction and dispersion of Fe species.

3.2.2. Catalytic performance of catalysts

The series of Fe/CNS(T) catalysts were applied to FTS reaction in order to investigate the influence of N species on catalytic performance. The results of activity and product selectivity at TOS of 30 h are summarized in Table 6. With increasing carbonization temperature, the CO conversion exhibits a ‘volcano’ trend, which yields an optimum (51.2%) at 1000 °C. However, it suddenly decreases to 31.0% as the carbonation temperature is elevated to 1100 °C. To alleviate the impact of conversion on product distribution, we regulated space velocity, keeping CO conversion at the same level. As listed in Table S3, the calcination temperature has little effect on product selectivity. Only the methane and lower olefins selectivity

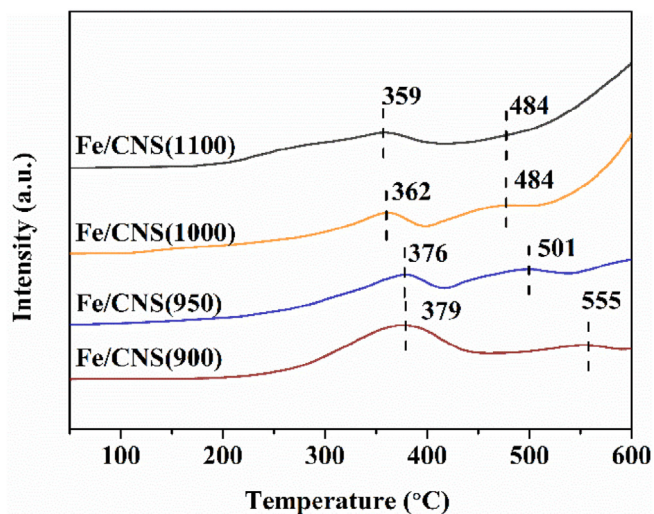


Fig. 11. H₂-TPR profiles of Fe/CNS(T) catalysts. (A colour version of this figure can be viewed online.)

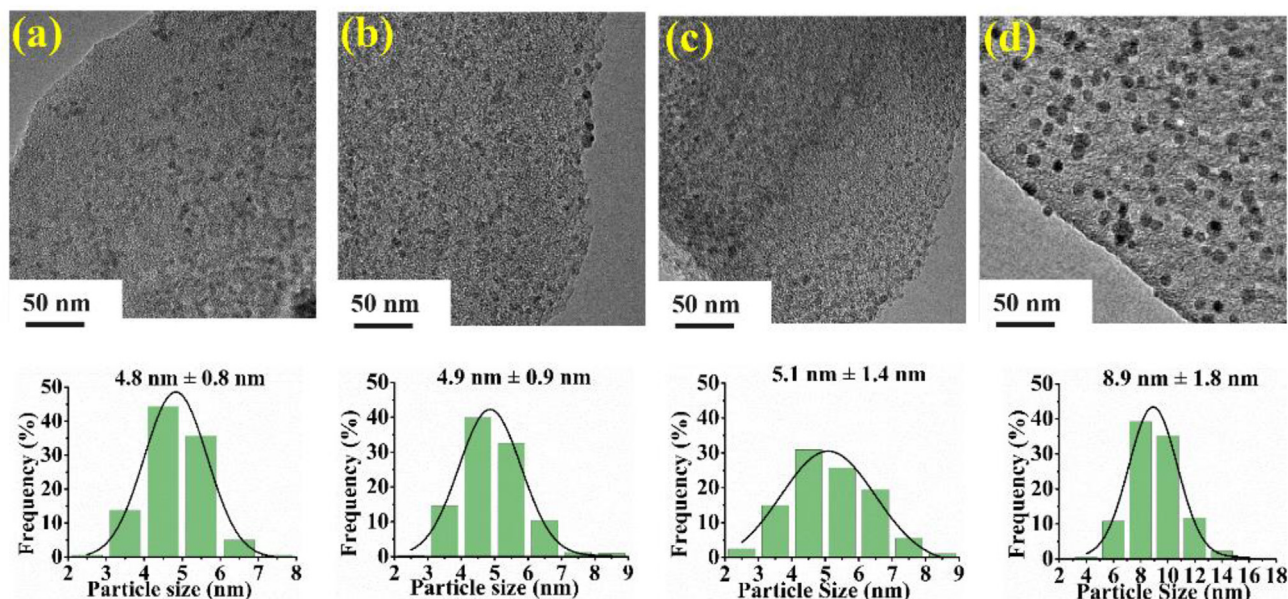


Fig. 12. TEM images of the Fe/CNS(T) catalysts: Fe/CNS(900) (a), Fe/CNS(950) (b), Fe/CNS(1000) (c) and Fe/CNS(1100) (d) (diagrams below them are the iron particle size distribution of corresponding catalysts). (A colour version of this figure can be viewed online.)

changes slightly.

High stability of Fe/CNS(T) was achieved even at 100 h on stream, as shown in Fig. S4. After reaction, Fe particles are surrounded by an amorphous carbonaceous layer, leading to an increase of particle size in Fig. S5. Nevertheless, the used catalysts are able to maintain uniform dispersion of Fe species. Therefore, we concluded that the N-anchoring effectively inhibits particle aggregation during reaction. The lattice fringes in Fig. S6 reveal the presence of χ -Fe₅C₂ and Fe₃O₄ in the used Fe/CNS(1000) catalysts, which is in agreement with other findings [16,51].

3.3. N-doping effect on FTS performance

Comparing with Raman spectra of the supports (Fig. 13a and b), the G bands for Fe/CNS(T) samples upshift with values of 3.1 cm⁻¹, 6.2 cm⁻¹, 9.3 cm⁻¹, and 3.1 cm⁻¹ respectively. The shift can be ascribed to electron transfer from the carbon support to the metal [54]. More importantly, the degree of G band shift is found to be consistent with the amount of pyrrolic N in Fe/CNS(T). Further comparison of Fe 3p XPS spectra also confirms the changes in the electronic structure of iron species for Fe/CNS(T) samples. In Fig. 13c, the binding energy of Fe 3p gradually shifts to lower values as the carbonization temperature increases from 900 to 1000 °C. However, for Fe/CNS(1100), the binding energy of Fe 3p shifts towards higher value of 55.7 eV, which indicates that the samples with high pyrrolic N content hold relatively lower Fe 3p BEs. Thus, it is speculated that pyrrolic N functions as the dominant electron

donor to Fe species. Table 7 presents phase composition of the used Fe/CNS(T) catalysts obtained from Mössbauer spectra (Fig. S8). The N content and nature effect Fe carburization and the Fe/CNS(1000) possesses the highest ratio of Fe₅C₂. The content of Fe₅C₂ is consistent with that of pyrrolic N content, resulting in changes in catalytic activity. Together with Raman, XPS and Mössbauer analysis, it is conceivable that pyrrolic N enriches the electron density of Fe species, consequently promoting Fe carburization and CO conversion.

4. Conclusion

In this work, carbon materials derived from ZIF-8 precursors with different morphologies, as well as a commercial AC, were investigated as supports in FTS. The well-defined structure of ZIF-8 favored high N dispersion in the derived carbons and the morphologies were maintained during pyrolysis even at high temperatures (900–1100 °C). Compared with N-free carbon (AC), incorporation of N into the carbon matrix (i.e. CP and CNS) achieved improved dispersion of Fe species, while the higher graphitization degree of monoclinic ZIF-8 (CNS) derivative further promoted electron transfer. Therefore, both morphology and N doping significantly affect the formation of Fe₅C₂. As a result, higher CO conversion was achieved and N-doping remarkably regulates the product distribution towards short-chain hydrocarbons. Further investigation on pyrolysis temperature of ZIF-8-NS suggested that the configuration and content of the N dopant can be precisely

Table 6
FTS performance of Fe/CNS(T) catalysts.

Catalysts	CO conv. (%)	FTY (μmol _{CO} g _{Fe} ⁻¹ s ⁻¹)	CO ₂ selectivity (%)	Hydrocarbon selectivity (C-mol%)					O/P
				CH ₄	C ₂ –C ₄	C ₂ –C ₄	C ₅ +	Oxy.	
Fe/CNS(900)	33.9	198	31.9	26.5	28.4	15.8	28.4	0.9	1.8
Fe/CNS(950)	40.2	233	32.2	24.3	23.2	15.9	35.6	1.0	1.5
Fe/CNS(1000)	51.2	295	33.0	23.6	22.0	17.2	36.3	0.9	1.3
Fe/CNS(1100)	31.0	183	31.2	28.4	23.0	16.3	31.6	0.7	1.4

Reaction conditions: 0.1 g cat., 340 °C, 1 MPa, H₂/CO = 1, GHSV = 9000 mL g⁻¹ h⁻¹, TOS = 30 h.

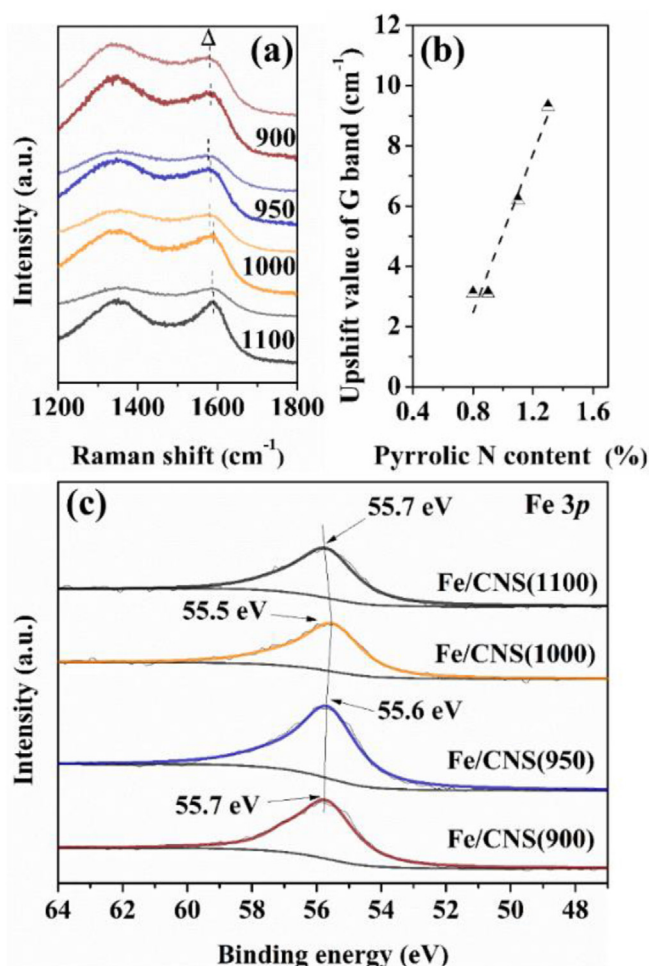


Fig. 13. Raman shift of CNS(T) (light colors) and Fe/CNS(T) (dark colors) (a); G-band upshift values from supports to catalysts vs. pyrrolic N contents (b); Fe 3p spectra of Fe/CNS(T) catalysts (c). (A colour version of this figure can be viewed online.)

Table 7

⁵⁷Fe Mössbauer parameters of the used Fe/CNS(900), Fe/CNS(1000) and Fe/CNS(1100) catalysts.

Used catalysts	IS (mm·s ⁻¹)	QS (mm·s ⁻¹)	H (T)	Phase ascription	A (%)
Fe/CNS(900)	0.34	0.85	—	Fe(II)/Fe(III)	44.3
	0.10	0.15	18.4	χ-Fe ₅ C ₂ (A)	21.7
	0.20	0.09	21.6	χ-Fe ₅ C ₂ (B)	21.7
	0.27	0.12	10.9	χ-Fe ₅ C ₂ (C)	12.3
	0.29	0.89	—	Fe(II)/Fe(III)	31.2
Fe/CNS(1000)	0.15	0.03	17.8	χ-Fe ₅ C ₂ (A)	24.4
	0.20	0.05	22.0	χ-Fe ₅ C ₂ (B)	24.5
	0.16	0.05	10.1	χ-Fe ₅ C ₂ (C)	19.9
	0.34	0.80	—	Fe(II)/Fe(III)	48.3
	0.17	0.03	18.0	χ-Fe ₅ C ₂ (A)	18.9
Fe/CNS(1100)	0.22	0.09	21.6	χ-Fe ₅ C ₂ (B)	18.9
	0.20	0.11	11.1	χ-Fe ₅ C ₂ (C)	13.9

regulated by changing temperature from 900 to 1100 °C. By quantifying the N with different configurations, we confirmed that pyrrolic N functioned as the main electron-donor to Fe species, which facilitated Fe carburization and contributed to CO conversion in FTS. Proceeding from MOF precursor, this work achieves controllable synthesis of N-dopant two-dimensional carbon materials and their catalytic application in FTS. By analyzing the structure-performance relationship, we provide an understanding

of N dopant effect on supported Fe catalysts, inspiring further development of metal supported carbon catalysts.

CRediT authorship contribution statement

Qiao Zhao: Methodology, Validation, Formal analysis, Investigation, Writing - original draft. **Shouying Huang:** Conceptualization, Supervision, Writing - review & editing, Funding acquisition. **Xiaoxue Han:** Writing - review & editing, Validation, Visualization. **Jiajia Chen:** Writing - review & editing. **Junhu Wang:** Mössbauer spectroscopy experiment and analysis. **Alexandre Rykov:** Mössbauer spectra analysis. **Yue Wang:** Writing - review & editing, Review & Editing. **Meiyan Wang:** Writing - review & editing, Review & Editing. **Jing Lv:** Writing - review & editing, Review & Editing, Funding acquisition. **Xinbin Ma:** Funding acquisition, and, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbon.2020.11.019>.

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Supporting Information

Highly active and controllable MOF-derived carbon nanosheets supported iron catalysts for Fischer-Tropsch synthesis

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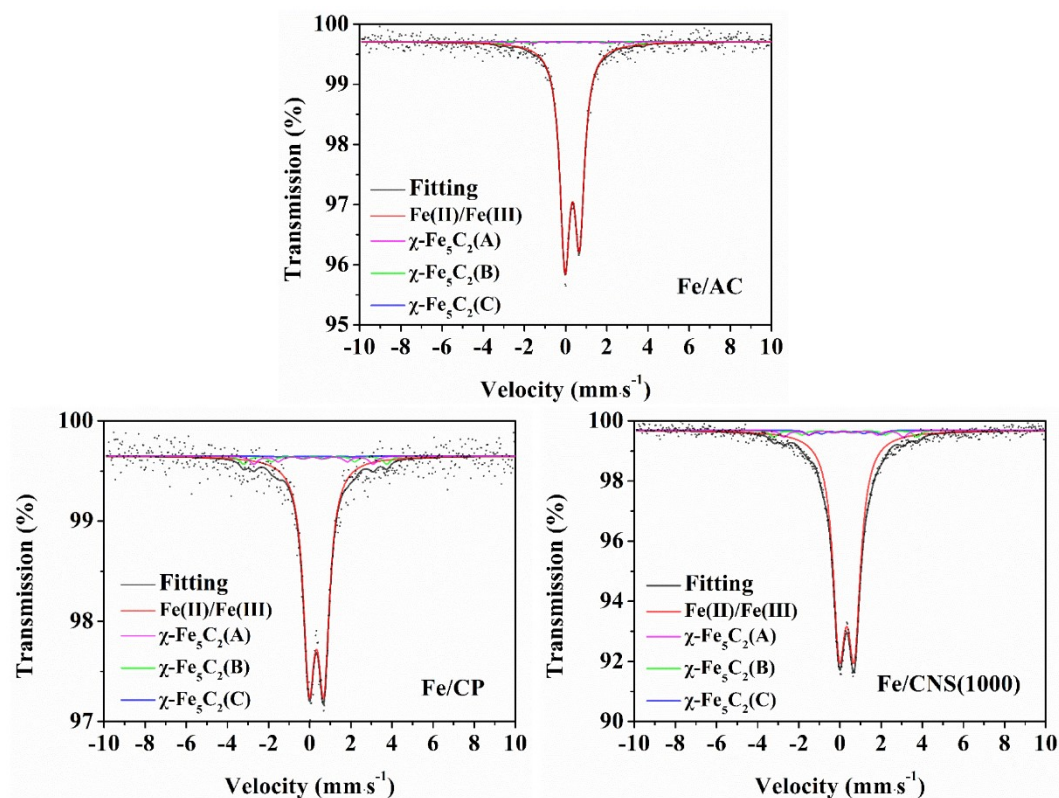


Fig. S1. ^{57}Fe Mössbauer spectra of the fresh Fe/AC, Fe/CP and Fe/CNS(1000) catalysts.

Table S1 ^{57}Fe Mössbauer parameters of the fresh Fe/AC, Fe/CP and Fe/CNS(1000) catalysts.

Fresh catalysts	IS ($\text{mm}\cdot\text{s}^{-1}$)	QS ($\text{mm}\cdot\text{s}^{-1}$)	H (T)	Phase ascription	A (%)
Fe/AC	0.32	0.70	-	Fe(II)/Fe(III)	97.2
	0.17	0.03	18.0	$\chi\text{-Fe}_5\text{C}_2(\text{A})$	1.1
	0.22	0.09	21.6	$\chi\text{-Fe}_5\text{C}_2(\text{B})$	1.1
	0.20	0.11	11.1	$\chi\text{-Fe}_5\text{C}_2(\text{C})$	0.6
Fe/CP	0.33	0.72	-	Fe(II)/Fe(III)	89.8
	0.17	0.03	18	$\chi\text{-Fe}_5\text{C}_2(\text{A})$	4.8
	0.22	0.09	21.6	$\chi\text{-Fe}_5\text{C}_2(\text{B})$	4.8
	0.20	0.11	11.1	$\chi\text{-Fe}_5\text{C}_2(\text{C})$	0.6
Fe/CNS(1000)	0.33	0.75	-	Fe(II)/Fe(III)	90.5
	0.17	0.03	18.0	$\chi\text{-Fe}_5\text{C}_2(\text{A})$	3.8
	0.22	0.09	21.6	$\chi\text{-Fe}_5\text{C}_2(\text{B})$	3.8
	0.20	0.11	10.9	$\chi\text{-Fe}_5\text{C}_2(\text{C})$	1.9

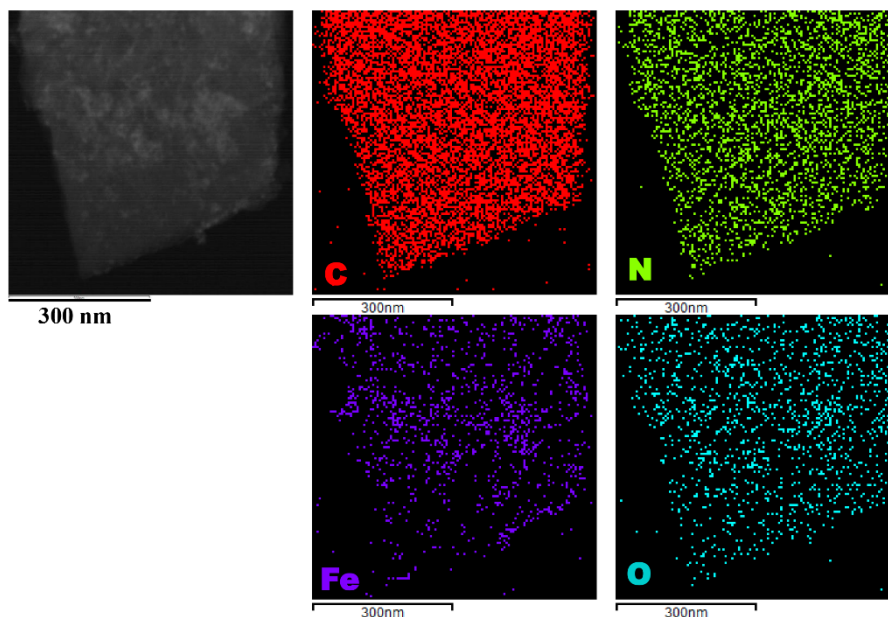


Fig. S2. TEM images of the Fe/CNS(1000) catalysts in dark field, and corresponding mappings of C, N, Fe and O.

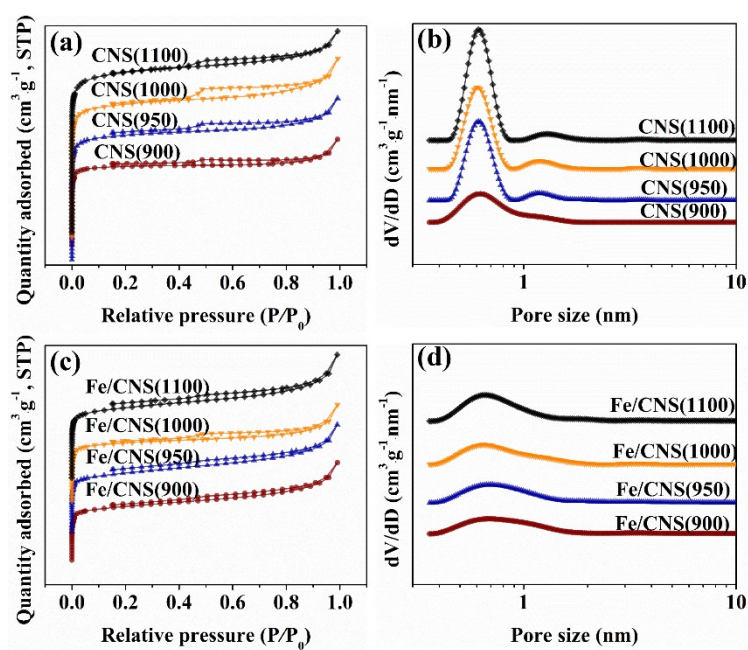


Fig. S3. N_2 adsorption-desorption isotherms (a, c) and pore size distribution (b, d) of the CNS(T), and corresponding iron-based catalysts.

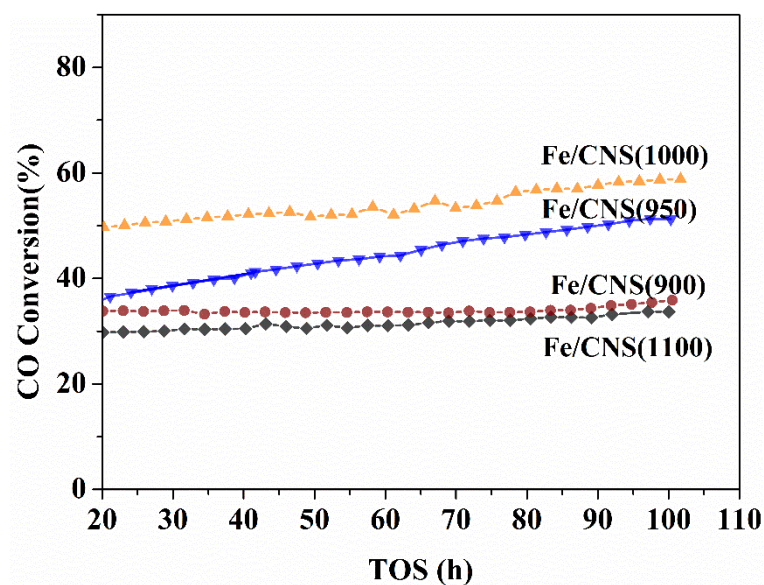
Table S2 Surface area and pore structure parameters of CNS(T) and Fe/CNS(T) catalysts.

samples	S^{BET} (m^2g^{-1})	V_{total} (cm^3g^{-1})	V_{micro} (cm^3g^{-1})	D_a (nm)
CNS(900)	272	0.12	0.10	1.7
CNS(950)	416	0.18	0.15	1.8
CNS(1000)	438	0.20	0.15	1.8
CNS(1100)	524	0.23	0.18	1.8
Fe/CNS(900)	194	0.12	0.06	2.3
Fe/CNS(950)	208	0.13	0.07	2.4
Fe/CNS(1000)	210	0.14	0.08	2.1
Fe/CNS(1100)	250	0.15	0.09	2.3

Table S3 FTS performance of Fe/CNS(T) catalysts at changeable space velocity.

Catalysts	CO conv. (%)	FTY ($\mu\text{mol}_{\text{CO}}$ $\text{g}_{\text{Fe}}^{-1}\text{s}^{-1}$)	CO ₂ selectivity (%)	Hydrocarbon selectivity (C-mol%)					O/P
				CH ₄	C ₂ -C ₄ ⁼	C ₂ -C ₄ ^o	C ₅ ⁺	Oxy.	
Fe/CNS(900)	33.9	198	31.9	26.5	28.4	15.8	28.4	0.9	1.8
Fe/CNS(950)	34.9	278	30.5	27.2	26.3	15.3	30.4	0.8	1.7
Fe/CNS(1000)	34.7	323	31.6	27.5	24.9	15.6	31.2	0.8	1.6
Fe/CNS(1100)	31.2	191	28.6	28.4	23.0	16.3	31.6	0.7	1.4

Reaction conditions: 0.1 g cat., 340 °C, 1 MPa, H₂/CO=1, GHSV=9000-14000 mL·g⁻¹·h⁻¹, TOS=30 h.

**Fig. S4.** CO conversion of Fe/CNS(T) catalysts on TOS. ($P = 1$ MPa, $\text{H}_2/\text{CO} = 1$, $\text{GHSV} = 9000 \text{ mL}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$, $T = 340$ °C)

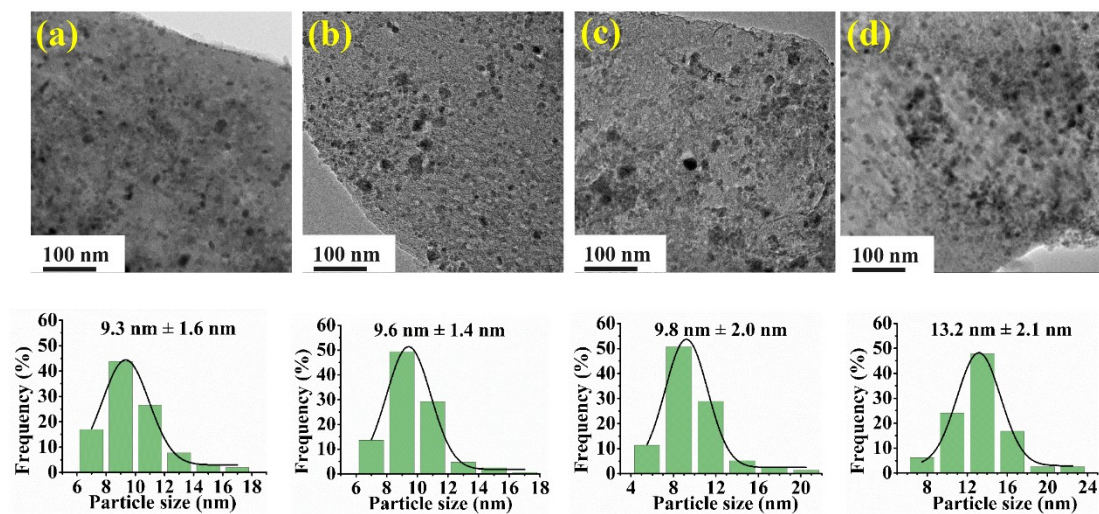


Fig. S5. TEM images of the used Fe/CNS(T) after 100 h reaction: Fe/CNS(900) (a), Fe/CNS(950) (b), Fe/CNS(1000) (c) and Fe/CNS(1100) (d) (diagrams below them are the iron particle size distribution of corresponding catalysts).

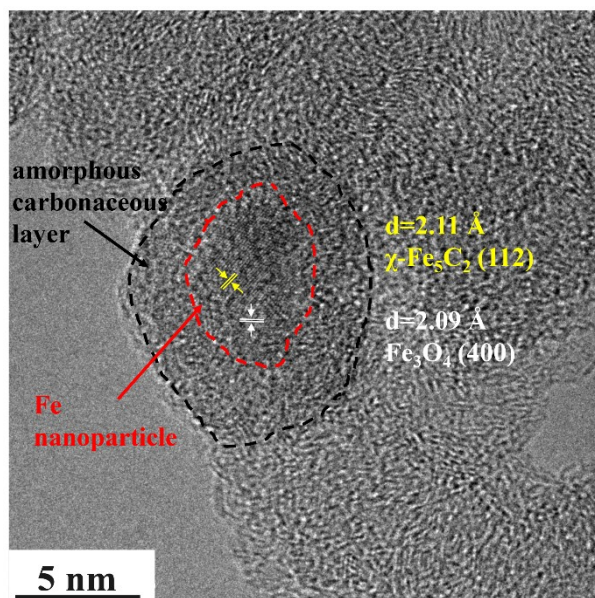


Fig. S6. HR-TEM images of the used Fe/CNS(1000).

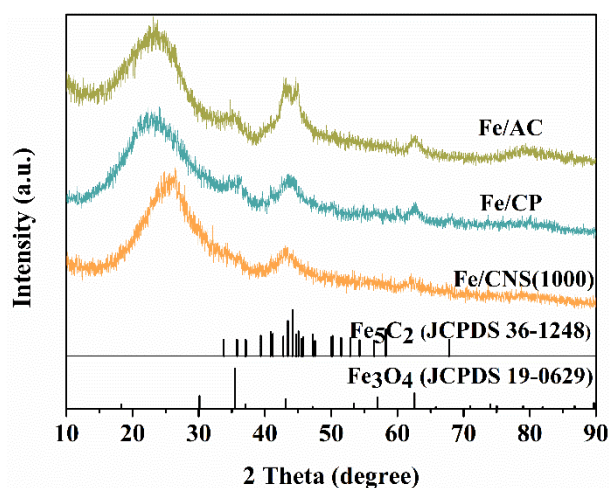


Fig. S7. XRD patterns of the used Fe/CNS(1000), Fe/CP, Fe/AC.

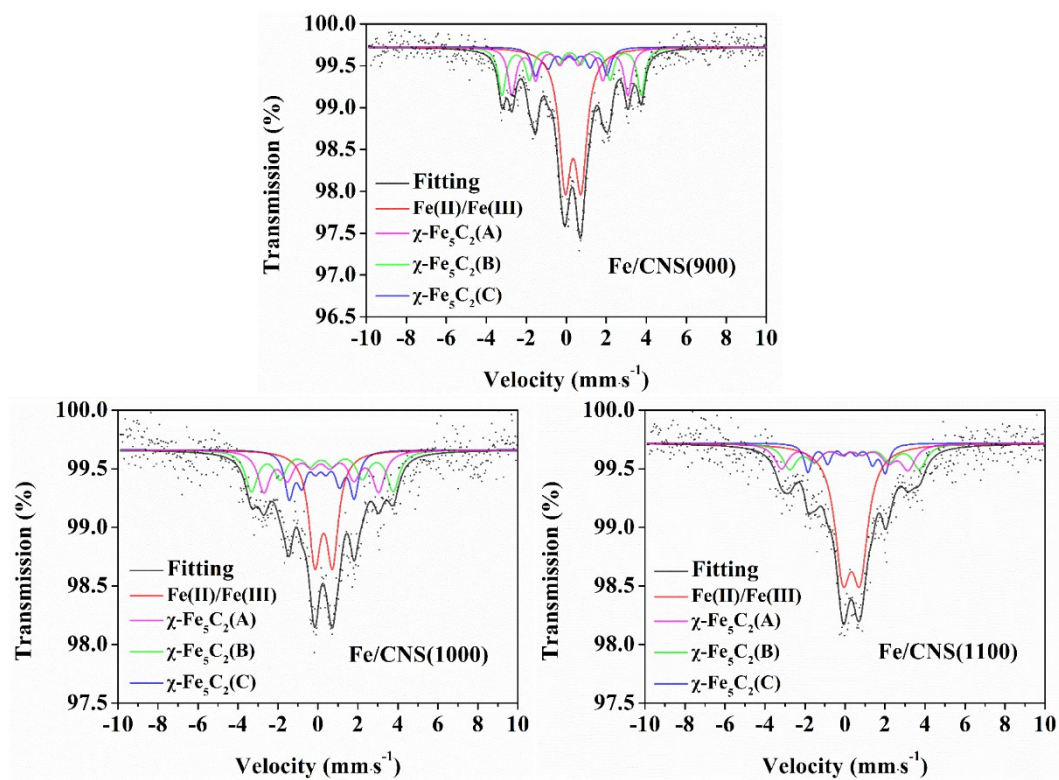


Fig. S8. ^{57}Fe Mössbauer spectra of the used Fe/CNS(900), Fe/CNS(1000) and Fe/CNS(1100) catalysts.